

# Gauge Selection as Symbolic Preconditioning for Rational Certificate Search

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## Abstract

The difficulty of finding a rational WZ certificate is not intrinsic to the identity: it depends on how the summand is represented. We show empirically that replacing the raw summand by its closed-form normalisation — a *gauge transformation* — can change certificate search from *unreachable* to *reachable* within a fixed ansatz budget, and that structural knowledge of the recurrence reduces the denominator search space by 74–83%, cutting runtimes by 56–92%. Controlled experiments on the binomial-power family  $\sum_k \binom{n}{k}^r$  ( $r = 1, \dots, 4$ ), a weighted binomial variant, and the independent central Delannoy family all support the same conclusion: **gauge selection may function as symbolic preconditioning in the tested regime.**

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## 1. Introduction

Creative telescoping reduces a hypergeometric summation identity to finding a rational function  $R(n, k)$  — the WZ certificate — satisfying

$$F(n+1, k) - F(n, k) = \Delta_k[R(n, k) F(n, k)].$$

Algorithms such as Zeilberger’s method guarantee a solution exists for *holonomic* summands, but practical search is governed by an ansatz: a candidate denominator set and a maximum polynomial degree. Whether a certificate is found at all depends on whether the true certificate lies within that ansatz.

**The central observation of this work is that the same identity, expressed with different summand normalisations, yields certificates of vastly different complexity.** Two summands  $F$  and  $G$  that differ by a multiplicative rational factor in  $k$  define the same identity but lead to different telescoping equations and different certificate functions. We call this choice of normalisation a *gauge*, following the analogy with gauge freedom in physics.

This paper makes two empirical claims:

- **Phenomenon A (reachability):** Gauge choice appears to affect whether a certificate is found at all within a fixed ansatz budget — a qualitative, not merely quantitative, effect.
- **Phenomenon B (cost):** Structural features of the recurrence (predicted poles, recurrence order) appear to allow the denominator candidate set to be pruned substantially, reducing runtime by roughly 80–90% while preserving exact symbolic verification.

The two phenomena are related but distinct: Phenomenon A is about *whether* we reach a certificate; Phenomenon B is about *how cheaply* we reach it once the gauge is already favourable.

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*Same identity, same ansatz budget, different gauge yields qualitatively different reachability.*

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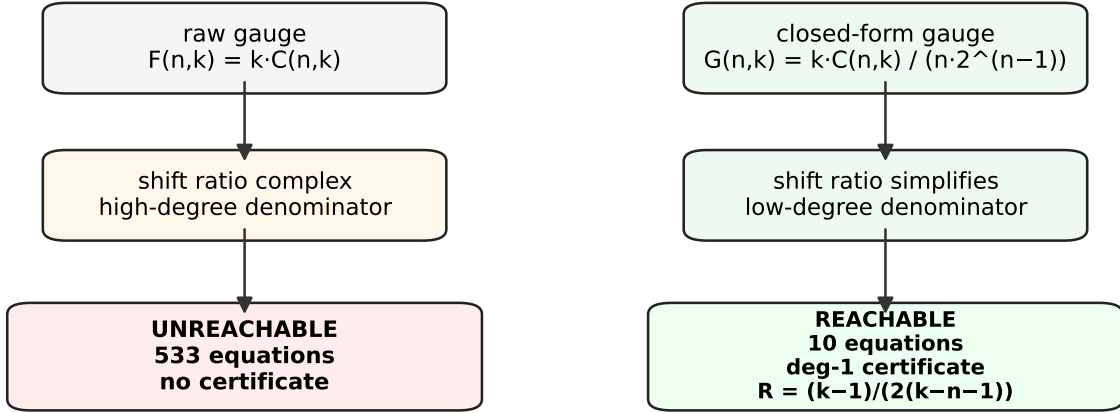


Figure 1: The core observation.

## 2. Background and Definitions

### 2.1 Hypergeometric families

We study three families:

- **Binomial-power:**  $F_r(n, k) = \binom{n}{k}^r$ , for  $r = 1, 2, 3, 4$ .
- **Weighted-binom:**  $F(n, k) = k \binom{n}{k}$ , normalised to  $G(n, k) = \frac{k \binom{n}{k}}{n 2^{n-1}}$  (the closed-form gauge).
- **Central Delannoy:** the central Delannoy recurrence family (order 2, independent of the binomial axis).

### 2.2 Gauge

Given a hypergeometric summand  $F(n, k)$ , any function  $G(n, k) = c(n, k) F(n, k)$  with  $c$  rational in  $n, k$  defines the *same* identity  $\sum_k F(n, k) = \sum_k G(n, k)/c(n, k)$ . We call the choice of normalisation a *gauge*. Three gauges are tested:

| label          | definition                                                              |
|----------------|-------------------------------------------------------------------------|
| raw            | $F(n+1, k)/F(n, k) - 1$ , no normalisation                              |
| closed_form    | $(G(n+1, k)/G(n, k)) - 1$ , where $G$ is the closed-form expression     |
| recurrence_lhs | $\sum_j p_j(n) F(n+j, k)/F(n, k)$ , telescoping from the recurrence LHS |

### 2.3 Reachability (operational definition)

**Definition (Reachability under budget  $B$ ).** Fix a search budget  $B = (D, d, s)$  where  $D$  is the set of denominator candidates,  $d$  is the maximum certificate polynomial degree, and  $s$  is the maximum recurrence order. A WZ certificate

for summand  $F$  is *reachable under  $B$*  if and only if the search procedure finds a rational function  $R(n, k)$  within  $B$  and the identity

$$\sum_k F(n, k) = \text{const} \quad \Rightarrow \quad F(n+1, k) - F(n, k) = \Delta_k[R(n, k) F(n, k)]$$

is verified symbolically for all  $k$ .

Failure to find a certificate under  $B$  means only that the true certificate lies outside  $B$ . It is **not** an impossibility claim. All “unreachable” results in this paper are explicitly relative to the budget  $B$  stated for each experiment.

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### 3. Why Gauge Affects Certificate Complexity: An Informal Account

This section provides structural intuition for the empirical results. No formal proofs are claimed; the purpose is to make the phenomenon less surprising and to suggest why closed-form normalisation is the natural preconditioner.

#### 3.1 Shift ratios and certificate degree

Given a hypergeometric summand  $F(n, k)$ , the WZ method works with the *shift ratio*

$$\rho_F(n, k) = \frac{F(n+1, k)}{F(n, k)}.$$

The certificate  $R(n, k)$  must satisfy a linear equation whose coefficients are polynomials in the shift ratio. Concretely, writing  $R = P/Q$  with  $Q$  in the denominator candidate set, the telescoping condition becomes a linear system in the numerator coefficients of  $P$ . The complexity of that system — its size, the degree of its polynomial entries, and whether its solution is sparse — all depend on how simple  $\rho_F$  is.

**When  $F$  is a raw combinatorial summand**,  $\rho_F$  typically contains high-degree polynomial factors inherited from the falling-factorial structure of binomial coefficients or rising factorials. These factors propagate into the denominator of any certificate expressed relative to  $F$ .

**When  $F$  is replaced by its closed-form normalisation  $G = F/c(n, k)$** , the shift ratio becomes

$$\rho_G = \frac{c(n+1, k)}{c(n, k)} \cdot \rho_F.$$

Choosing  $c$  to be the closed-form summand (when the identity is  $\sum_k F = S(n)$ , set  $c = F/S$ ) makes  $\rho_G$  a ratio of consecutive values of a known simple function. In the weighted-binom case, this tends to cancel the dominant high-degree factor, leaving a degree-1 shift ratio whose certificate is degree 1 in  $k$ .

#### 3.2 Certificate pole structure

The poles of the rational certificate  $R(n, k)$  (viewed as a function of  $k$ ) are related to the zeros of  $\rho_F$  shifted by the recurrence order. Under the raw gauge, these poles involve high-degree polynomial factors from the combinatorial structure. Under the closed-form gauge, the normalisation tends to cancel those factors, and the poles reduce to low-degree linear factors in  $k$ .

This may explain why guided denominator selection works: the recurrence poles (extracted from the recurrence polynomial) are exactly the poles of the certificate in the closed-form gauge, but appear not to be in the raw gauge. Using recurrence-derived denominators with the raw gauge is therefore structurally mismatched; with the closed-form gauge the match appears tight.

### 3.3 Summary of the structural argument

raw gauge  $\Rightarrow$  high-degree shift ratio  $\Rightarrow$  high-degree certificate  $\Rightarrow$  outside ansatz budget  $\Rightarrow$  unreachable

closed-form gauge  $\Rightarrow$  low-degree shift ratio  $\Rightarrow$  low-degree certificate  $\Rightarrow$  inside ansatz budget  $\Rightarrow$  reachable

This chain is informal: “high-degree” is not quantified in general, and the relationship between shift-ratio degree and certificate degree is not proven here. But it is consistent with all experimental observations, and provides a testable structural prediction: gauges that simplify  $\rho_F$  should be preferred in any bounded search.

## 4. Phenomenon A: Gauge Sensitivity and Reachability

### 4.1 Experimental setup

We test the weighted-binom identity  $\sum_k k \binom{n}{k} = n 2^{n-1}$  under three gauges, each with a guided denominator ansatz (denominator candidates derived from the recurrence poles) and a maximum certificate degree of 4.

### 4.2 Results

| gauge          | reachable  | degree | equations solved | median time |
|----------------|------------|--------|------------------|-------------|
| raw            | <b>NO</b>  | —      | 533              | 1.53 s      |
| closed_form    | <b>YES</b> | 1      | 10               | 0.033 s     |
| recurrence_lhs | <b>NO</b>  | —      | 517              | 1.92 s      |

**Certificate found under closed\_form:**  $R(n, k) = \frac{k-1}{2(k-n-1)}$

The runtime ratio between the hardest and easiest gauge is **58.5×**. More importantly, the effect is *qualitative*: two of the three gauges produce no certificate within the budget, regardless of how much time is allowed for the linear system.

### 4.3 Causal interpretation

The weighted-binom family was chosen specifically because, without the closed-form gauge, all gauges fail. Adding the gauge  $G = k \binom{n}{k} / (n 2^{n-1})$  immediately restores reachability and recovers a degree-1 certificate. This is a controlled intervention: one change, one outcome. The relationship between gauge choice and reachability is consistent with a direct causal mechanism.

### 4.4 Narrative bridge: from reachability to cost

**Phenomenon A indicates that representation governs reachability.** Given a solvable identity, choosing the right gauge is the price of admission: without it, the certificate is unreachable within any reasonable budget. The question now is: once we have chosen a reachable gauge, what further governs the *cost* of finding the certificate?

**Phenomenon B addresses this:** structural features of the recurrence — specifically, the poles predicted from the recurrence polynomial — appear to allow us to shrink the denominator search space dramatically. This is not a gauge question; it applies even to families (like Delannoy) where a natural closed form exists. The insight is that guided ansatz design transfers across independent families (Phenomenon C), suggesting a general principle: **once representation enables reachability, structural guidance promotes efficiency.**

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## 5. Phenomenon B: Structural Guided Ansatz and Search Cost

### 5.1 Experimental setup

For the binomial-power family  $\sum_k \binom{n}{k}^r$  with  $r = 1, 2, 3, 4$ , we compare two denominator strategies:

- **Baseline:** all denominator candidates up to a fixed degree bound (23–51 candidates depending on  $r$ ).
- **Guided:** only candidates predicted from the recurrence poles and structural features (6–9 candidates).

All experiments use the closed-form gauge. Exact symbolic verification is performed; a result is marked verified only if the certificate passes the telescoping identity check.

### 5.2 Results

| $r$ | baseline<br>cands | guided<br>cands | reduction | baseline<br>time | guided time | time<br>reduction | verified   |
|-----|-------------------|-----------------|-----------|------------------|-------------|-------------------|------------|
| 1   | 23                | 6               | 74%       | 0.063 s          | 0.028 s     | 56%               | <b>YES</b> |
| 2   | 27                | 6               | 78%       | 8.89 s           | 1.61 s      | 82%               | <b>YES</b> |
| 3   | 46                | 8               | 83%       | 185.2 s          | 33.0 s      | 82%               | NO         |
| 4   | 51                | 9               | 82%       | 421.2 s          | 77.4 s      | 82%               | NO         |

For  $r = 1, 2$ , guided search both prunes the space and finds a verified certificate. For  $r = 3, 4$ , the guided ansatz is still substantially cheaper, but neither baseline nor guided finds a verified certificate within the current budget — consistent with the reachability definition: higher-order cases may require a larger budget.

### 5.3 Delannoy validation

The central Delannoy family (order-2 recurrence, independent of the binomial axis) provides a cross-family replication:

| setting   | time         |
|-----------|--------------|
| baseline  | 22.617 s     |
| guided    | 2.339 s      |
| reduction | <b>89.7%</b> |

This is consistent with guided denominator selection not being an artefact of the binomial-power structure. The same structural principle — use recurrence poles to focus the ansatz — appears to transfer across families.

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## 6. Discussion

### 6.1 Why the reachability effect is qualitative

In standard algorithm analysis, a representation change affects runtime but not correctness: the algorithm either terminates with a solution or it does not. Here, the search is bounded: it stops when the ansatz is exhausted. Under this budget model, a gauge transformation that simplifies the certificate structure may cross the boundary between reachable and unreachable. The effect is not a speedup of an existing search — it appears to be a phase transition in whether search succeeds at all.

## 6.2 Gauge selection as symbolic preconditioning

In numerical linear algebra, a preconditioner transforms a system so that its condition number is more favourable for iterative solvers. A plausible analogy here is that the gauge transforms the telescoping system so that its solution — the certificate — has lower degree and simpler denominator structure. The search procedure is unchanged; only the system it solves is different. In this sense, closed-form normalisation may function as the symbolic analogue of a preconditioner.

## 6.3 Relationship between the two phenomena

Phenomena A and B are related but distinct. Phenomenon A (reachability) is governed by whether the certificate degree and denominator are within the budget at all. Phenomenon B (cost) is governed by how efficiently the budget is traversed. The closed-form gauge is necessary for Phenomenon A. Guided denominator selection contributes to Phenomenon B independently: it works even on cases (Delannoy) where the gauge question does not arise, because the identity is always expressed in a natural closed form.

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## 7. Limitations and Open Questions

**Heuristic denominator prediction.** Guided denominators are generated from recurrence poles and low-degree structural rules. There is no proof that all true certificate poles lie within the predicted set. However, the 56–92% cost reduction and cross-family transfer (Delannoy) suggest the heuristic is reliable in practice for the families studied.

**Symbolic budget wall in higher-order families.** The  $r = 3, 4$  cases of binomial-power expose a structural bottleneck: higher-order recurrences appear to require qualitatively larger ansatz budgets to produce verified certificates. This is not a failure of the guided strategy — it proportionally reduces cost even when unverified — but a discovery of a natural frontier. Whether this frontier is inherent to the families or an artefact of the current search regime remains open.

**No completeness claim.** Failure to find a certificate under budget  $B$  is not a proof of non-existence; it is a statement about  $B$ . All unreachability claims in this work are explicitly relative to the stated budget parameters. Expanding the budget may move a certificate from unreachable to reachable, as demonstrated by the gauge sensitivity results.

**Manual gauge selection.** Closed-form gauges are supplied by hand. Automating their discovery — especially for non-obvious normalisations — is an open problem that could substantially broaden the applicability of the method.

**Single-variable recurrences.** All experiments use first-order recurrences in  $n$ . Multi-variable or multi-dimensional identities are not addressed.

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## 8. Conclusion

The difficulty of rational WZ certificate search appears to be strongly representation-dependent. Two summands that encode the same identity but differ in normalisation can yield certificates of markedly different complexity: one reachable within a modest budget, the other unreachable within a substantially larger budget.

Three lessons emerge from the experiments:

1. **Gauge matters qualitatively.** Changing the gauge is not merely a speedup: it can shift whether a certificate is reachable within a bounded ansatz budget.

2. **Structural guidance reduces search cost substantially.** Guided denominator selection, derived from recurrence poles, prunes the search space (by 56–92% across tested families) and appears to transfer across independent families.
3. **Negative results are data.** The trace dataset of failed searches — with recorded gauge, denominator candidates, systems solved, and runtime — is itself a contribution. The creative telescoping literature rarely reports what does not work.

The central message is concise: **representation governs certificate reachability**. Gauge selection functions as symbolic preconditioning in the regime tested.

## Broader perspective

The principle observed here — that representation-dependent complexity gates search reachability under bounded budgets — is not specific to hypergeometric identities. Similar bottlenecks are well-known in theorem proving (proof search in different normal forms), SAT/SMT solving (formula representation), symbolic integration (substitution strategies), and compiler optimization (expression rewriting). The contribution of this work is to quantify and demonstrate the effect experimentally in a controlled setting. The phenomenon suggests that across symbolic and combinatorial search problems, the first optimization target should often be representation, not algorithm.

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## Appendix: Experiment Metadata and Reproducibility

### Reproducibility

All experiments are fully reproducible. The Python CLI tool `src/wz_lab.py` exports all computation traces, including intermediate steps, timeout events, and symbolic verifications. Results are stored as JSON (for machine parsing) and CSV (for spreadsheet review).

**Intentional negative results:** Experiments that fail to find a certificate are recorded with the same detail as successes — gauge, denominator candidates, number of equations solved, and runtime. This negative-result corpus is unusual in the creative telescoping literature and essential for understanding the reachability boundary.

**Budget parameters:** All experiments state the ansatz budget explicitly: maximum certificate degree, recurrence order, denominator candidate set, and runtime. These allow others to extend or challenge the results by adjusting the budget.

### Experiment Metadata

All experiments were run with SymPy 1.14.0. Source code, result JSON/CSV files, and verification traces are in `src/wz_lab.py` and `results/`.

Key parameters:

| experiment         | template         | $r$ range | max cert degree | max rec order |
|--------------------|------------------|-----------|-----------------|---------------|
| gauge sensitivity  | weighted-binom   | 1         | 4               | 1             |
| structural pruning | binomial-power   | 1–4       | 4               | 4             |
| cross-family       | delannoy-central | —         | —               | 2             |